

PUKE 1.6. Series

sequence = an ordered set of numbers e.g.: $a_1, a_2, \dots, a_k, \dots, a_n$
 $1, 4, 7, 10, \dots$

finite

- first term ✓
- last term ✓

infinite

- continues forever

1. Binomial expansion of $(a+b)^n$

"two terms"

$$\begin{aligned}(a+b)^0 &= 1 \\ (a+b)^1 &= 1a + 1b \\ (a+b)^2 &= 1a^2 + 2ab + 1b^2 \\ (a+b)^3 &= 1a^3 + 3a^2b + 3ab^2 + 1b^3 \\ (a+b)^4 &= 1a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + 1b^4\end{aligned}$$

e.g.: $1b^4$
 index
 coefficient

$n=4$

$$\begin{array}{ccccccc} & & -1 & & -1 & & -1 & & -1 \\ & & 4 & & 3 & & 2 & & 1 & & 0 \\ \cdot & a & & & & & & & & & \\ & & +1 & & +1 & & +1 & & +1 \\ & & 0 & & 1 & & 2 & & 3 & & 4 \\ \cdot & b & & & & & & & & & \\ & & 4 & & 4 & & 4 & & \dots \end{array}$$

all terms have a total index of 4

$$\begin{array}{ccccccc} n=0: & & & & 1 & & \\ n=1: & & & 1 & & 1 & \\ n=2: & & 1 & & 2 & & 1 \\ n=3: & 1 & & 3 & & 3 & & 1 \\ n=4: & 1 & 4 & & 6 & & 4 & 1 \end{array}$$

The next row is then:

$$n=5: 1 \quad 5 \quad 10 \quad 10 \quad 5 \quad 1$$

= coefficients
 form a pattern:
 "Pascal's triangle"

Method 1.

$$\Rightarrow 1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5$$

Method 2

r $\rightarrow$ nC_r function on calculator
 = position

(more on this in Stats 5.2. Combinations)

$$\text{OR } \binom{n}{r} = \frac{n!}{r!(n-r)!}$$

e.g.: $(a+b)^5$. Find coefficient for $(2+1)^{\text{th}}$ term

$\rightarrow$ calculator $\binom{5}{2}$

$$1a^5 + 5a^4b + 10a^3b^2 + 10a^2b^3 + 5ab^4 + 1b^5$$

1st term always starts at 0

$$\binom{5}{0} \quad \binom{5}{1}$$

coefficient = 5 $\binom{5}{5}$

2. Binomial coefficients

= coefficients in expansion of $(1+x)^n$

↖ better way to deal with bigger numbers

* **Binomial theorem**: (n is a positive integer)

$$(1+x)^n = \binom{n}{0} + \binom{n}{1}x + \binom{n}{2}x^2 + \dots + \binom{n}{n}x^n, \text{ where } (r+1)^{\text{th}} \text{ term} = \binom{n}{r}x^r$$

or extend this rule to give:

$$(a+b)^n = \binom{n}{0}a^n + \binom{n}{1}a^{n-1}b^1 + \binom{n}{2}a^{n-2}b^2 + \dots + \binom{n}{n}b^n, \text{ where } (r+1)^{\text{th}} \text{ term} = \binom{n}{r}a^{n-r}b^r$$

$$\ast \binom{n}{0} = 1 ; \text{ e.g. } \binom{5}{0} = {}^5C_0 = \frac{5!}{0!5!} = \frac{5!}{0} = 1$$

$$\binom{n}{n} = 1 ; \text{ e.g. } \binom{5}{5} = {}^5C_5 = \frac{5!}{5!0!} = \frac{5!}{0} = 1$$

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{5 \times 4 \times \cancel{3!}}{\cancel{3!}2!} = \frac{5 \times 4}{2 \times 1} = 10$$

$$\binom{8}{4} = \frac{8!}{4!4!} = \frac{8 \times 7 \times 6 \times 5 \times \cancel{4!}}{4! \cancel{4!}} = \frac{8 \times 7 \times 6 \times 5}{4 \times 3 \times 2 \times 1} = 70$$

$$\hookrightarrow \binom{n}{r} = \frac{n \times (n-1) \times (n-2) \times \dots \times (n-r+1)}{r \times (r-1) \times (r-2) \times \dots \times 3 \times 2 \times 1}$$

3. Arithmetic progressions

= linear sequence + + +

term	1 st	2 nd	3 rd	4 th	5 th
	1	4	7	10	13
	a	$a+d$	$a+2d$	$a+3d$	$a+4d$

$\xrightarrow{+3}$ $\xrightarrow{+3}$ $\xrightarrow{+3}$ $\xrightarrow{+3}$

$$\hookrightarrow \boxed{n^{\text{th}} \text{ term} = a + (n-1)d}$$

↑ first term
← common difference

$l = \text{last term}$

* Σ all terms in a sequence = a "series"

* fear not, for all these formulas are provided on your exams :)

$$S_n = \frac{n}{2} (a+l)$$

or

$$S_n = \frac{n}{2} [2a + (n-1)d]$$

e.g.: $1 + 2 + 3 + \dots + 98 + 99 + 100 = ?$

$$= \frac{100}{2} \times (1+100) = 50 \times 101 = 5050$$



$$n^{\text{th}} \text{ term} = S_n - S_{n-1}$$

4. Geometric progressions

term

$$\begin{array}{cccc} 1^{\text{st}} & 2^{\text{nd}} & 3^{\text{rd}} & 4^{\text{th}} \\ 2 & 6 & 18 & 54 \\ a & ar & ar^2 & ar^3 \end{array}$$

x3 x3 x3

common ratio

$$n^{\text{th}} \text{ term} = ar^{n-1}$$

Σ geometric progression

$$S_n = \frac{a(1-r^n)}{1-r}$$

or

$$S_n = \frac{a(r^n-1)}{r-1}$$

e.g.: number series $3 + 6 + 12 + 24 + \dots$

Find sum of first 12 terms:

$$a=3$$

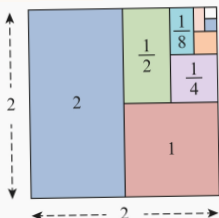
$$r=2$$

$$n=12$$

$$S_n = \frac{a(1-r^n)}{1-r} \Rightarrow S_{12} = \frac{3(1-2^{12})}{1-2} = 12\,285$$

5. Infinite geometric series

= terms continue forever



e.g.: $a=2$, $r=\frac{1}{2}$ } $2, 1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$

sum $S_1=2, S_2=3, S_3=3\frac{1}{2}, S_4=3\frac{3}{4}, S_5=3\frac{7}{8}, \dots$

↳ gets closer to 4 ; "series converges to 4"
= a convergent series

$$S_{\infty} = \frac{a}{1-r} \text{ provided that } -1 < r < 1$$

